ASSIGNMENT - 01

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MAT 2305

REAL ANALYSIS - II

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(01)(i)

i) $\sum_{n=1}^{\alpha} \sqrt{\frac{n}{2(n+1)}}$

1 - 50,
2 - 29
3 - 30
4 - 24
5 - 20
6 - 23
7 - 10
9 - 14
10 - 50

Let $a_n = \sqrt{\frac{n}{2(n+1)}}$ $\forall n \in \mathbb{N}$

$$\lim_{n\to\alpha} a_n = \lim_{n\to\alpha} \sqrt{\frac{n}{2(n+1)}}$$

$$= \lim_{n\to\alpha} \sqrt{\frac{n}{2n+2}}$$

$$= \lim_{n\to\alpha} \sqrt{\frac{\cancel{n}}{\cancel{n}[2+2/n]}}$$

$$= \lim_{n\to\alpha} \sqrt{\frac{1}{2}}$$

$$\lim_{n\to\alpha} a_n = \sqrt{\frac{1}{2}}$$

Since, $\lim_{n\to\alpha} a_n \neq 0$, the Series $\sum_{n=1}^{\alpha} \sqrt{\frac{n}{2(n+1)}}$ is divergent.

ii) $\sum_{n=1}^{\alpha} \cos(1/n^2)$

Let $a_n = \cos(1/n^2)$ $\forall n \in \mathbb{N}$

$$\lim_{n\to\alpha} a_n = \lim_{n\to\alpha} \cos(1/n^2)$$

$$= \lim_{n\to\alpha} \cos 0$$

$$= 1$$

$$\lim_{n\to\alpha} a_n = 1$$

Since, $\lim_{n\to\alpha} a_n \neq 0$, The Series $\sum_{n=1}^{\alpha} \cos(1/n^2)$ is divergent.

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(iii) $\sum_{n=1}^{\infty} \sqrt{\frac{3n^2+5n+4}{4n^2+4}}$

Let $a_n = \sqrt{\frac{3n^2+5n+4}{4n^2+4}}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \sqrt{\frac{3n^2+5n+4}{4n^2+4}}$$

$$= \lim_{n\to\infty} \sqrt{\frac{\cancel{n^2}\left(3+5/n+4/n^2\right)}{\cancel{n^2}\left[4+4/n^2\right]}}$$

$$= \lim_{n\to\infty} \sqrt{\frac{3}{4}}$$

$$= \frac{\sqrt{3}}{2}$$

$$\lim_{n\to\infty} a_n = \frac{\sqrt{3}}{2}$$

Since, $\lim_{n\to\infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \sqrt{\frac{3n^2+5n+4}{4n^2+4}}$ is divergent.

(iv) $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$

Let $a_n = \left(\frac{1}{n}\right)^{1/n} \quad \forall n \in \mathbb{N}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{1}{n}\right)^{1/n}$$

$$= \lim_{n\to\infty} \frac{1}{n^{1/n}}$$

$$= \lim_{n\to\infty} 1$$

$$= 1$$

$$\lim_{n\to\infty} a_n = 1$$

Since, $\lim_{n\to\infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$ is divergent.

(v) $\sum_{n=1}^{\infty} \left[\frac{2}{(-1)^n - 3}\right]^n$

Let $a_n = \left[\frac{2}{(-1)^n - 3}\right]^n \quad \forall n \in \mathbb{N}$

$$|a_n| = \left[\frac{2}{(-1)^n - 3}\right]^n$$

$$|a_n|^{1/n} = \frac{2}{(-1)^n - 3} \quad \forall n \in \mathbb{N}$$

$$|a_n|^{1/n} = \begin{cases} \frac{-2}{2} = -1 & ; \text{ if } n \text{ is even} \\ \frac{2}{-4} = -\frac{1}{2} & ; \text{ if } n \text{ is odd.} \end{cases}$$

$$A = \{|a_1|^{1/n}, |a_2|^{1/n}, |a_3|^{1/n}, \ldots, |a_n|^{1/n}, \ldots\}$$

$$A = \{-\tfrac{1}{2}, -1, -\tfrac{1}{2}, \ldots\}$$

$$\text{Sup } A = -\tfrac{1}{2}$$

$$\alpha = \lim \text{Sup } |a_n|^{1/n}$$

$$\alpha = \lim_{n \to \alpha} \left[\sup_{k \geq n} |a_k|^{1/k}\right]$$

$$\alpha = \lim_{n \to \alpha} -\tfrac{1}{2}$$

$$\alpha = -\tfrac{1}{2} < 1$$

By the root test, the series is absolutely Convergent.

$\therefore$ It is Convergent //

(02) (i) $\sum_{n=1}^{\alpha} \frac{1}{n^2+25}$

Let $a_n = \frac{1}{n^2+25}$ $\forall n \in \mathbb{N}$

$$\underset{\substack{\uparrow \\ a_n}}{\frac{1}{n^2+25}} < \underset{\substack{\uparrow \\ b_n}}{\frac{1}{n^2}}$$

$\sum_{n=1}^{\alpha} b_n = \sum_{n=1}^{\alpha} \frac{1}{n^2}$

using the p - Series test

$p = 2 > 1$

$\Rightarrow \sum_{n=1}^{\alpha} b_n = \sum_{n=1}^{\alpha} \frac{1}{n^2}$ is Convergent.

$\therefore \sum_{n=1}^{\alpha} b_n$ is Convergent.

$\Rightarrow \sum_{n=1}^{\alpha} a_n$ is Convergent by Comparison test

$\therefore \sum_{n=1}^{\alpha} \frac{1}{n^2+25}$ is Convergent.

(iii) $\sum_{n=1}^{\alpha} \frac{1}{n^2} \sin(1/n^2)$

Let $a_n = \frac{1}{n^2} \sin(1/n^2)$ $\forall n \in \mathbb{N}$

$$\underset{\substack{\uparrow \\ a_n}}{\frac{1}{n^2} \sin(1/n^2)} \leq \underset{\substack{\uparrow \\ b_n}}{\frac{1}{n^2}}$$

$\sum_{n=1}^{\alpha} b_n = \sum_{n=1}^{\alpha} \frac{1}{n^2}$.

using p - Series test;

$p = 2 > 1$

$\Rightarrow \sum_{n=1}^{\alpha} b_n = \sum_{n=1}^{\alpha} \frac{1}{n^2}$ is Convergent.

$\therefore \sum_{n=1}^{\alpha} a_n$ is Convergent by Comparison test. $\therefore \sum_{n=1}^{\alpha} 1/n^2 S$ is Convergent //

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(iv) $\sum_{n=1}^{\alpha} \frac{n}{n^2+3}$

(4)

Let $a_n = \frac{n}{n^2+3}$ $\forall n \in N$

$$\frac{n}{n^2+3} < \frac{n}{n^2} = \frac{1}{n}$$

$\uparrow$ a_n $\qquad$ $\uparrow$ b_n.

Now $l = \lim_{n\to\alpha} \frac{a_n}{b_n}$

$$= \lim_{n\to\alpha} \left[\frac{n}{n^2+3} \times \frac{n}{1}\right]$$

$$= \lim_{n\to\alpha} \frac{n^2}{n^2+3}$$

$$= \lim_{n\to\alpha} \frac{\cancel{n^2}\,1}{\cancel{n^2}[1+3/n^2]}$$

$l = 1 \in (0, \alpha)$

by which test

Also we know that $\sum_{n=1}^{\alpha} 1/n$ is divergent. $\therefore$ by the limit form of comparison test the series $\sum_{n=1}^{\alpha} \frac{n}{n^2+3}$ is divergent.

(03) (i) $\sum_{n=1}^{\alpha} \frac{\lfloor n}{n^n}$

we apply the limit form of the ratio test;

$a_n = \frac{\lfloor n}{n^n}$

$a_{n+1} = \frac{\lfloor n+1}{(n+1)^{n+1}}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\lfloor n+1}{(n+1)^{n+1}} \times \frac{n^n}{\lfloor n}\right|$$

$$= \left|\frac{(n+1)\,\cancel{n!}}{(n+1)^{n+1}} \times \frac{n^n}{\cancel{n!}}\right|$$

$$= \left|\frac{n^n (n+1)}{(n+1)^{n+1}}\right|$$

$$= \left|n^n (n+1)^{\cancel{1}-n-\cancel{1}}\right|$$

$$= \left|(n+1)^{-n}\, n^n\right|$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{n^n}{(n+1)^n}\right|$$

$$l = \lim_{n\to\alpha} \left|\frac{a_{n+1}}{a_n}\right|$$

$$= \lim_{n\to\alpha} \frac{n^n}{(n+1)^n}$$

$$= \lim_{n\to\alpha} \left(\frac{\cancel{n}}{\cancel{n}(1+1/n)}\right)^n$$

$$= \lim_{n\to\alpha} \left(\frac{1}{1+1/n}\right)^n$$

$$= \lim_{n\to\alpha} \frac{1}{(1+1/n)^n}$$

$$= \frac{1}{e} < 1$$

∴ But the limit form of the ratio test, the Series is absolutely Convergent.

∴ It is Convergent //.

(ii) $\sum_{n=1}^{\infty} \frac{n!}{3.5.7...(2n+1)}$

we apply the limit form of the ratio test;

Let $a_n = \frac{n!}{3.5.7...(2n+1)}$ $\forall n \in N$

$$a_{n+1} = \frac{(n+1)!}{3.5.7.....(2n+1)(2n+3)}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)!}{3.5.7....\cancel{(2n+1)}(2n+3)} \times \frac{3.5.7...\cancel{(2n+1)}}{n!}\right|$$

$$= \left|\frac{(n+1)\cancel{n!}}{(2n+3)\cancel{n!}}\right|$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)}{2n+3}\right|$$

$$\therefore l = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

$$= \lim_{n\to\infty} \left(\frac{n+1}{2n+3}\right)$$

$$= \lim_{n\to\infty} \frac{\cancel{n}(1+1/n)}{\cancel{n}(2+3/n)}$$

$$= \lim_{n\to\infty} 1/2$$

$$l = 1/2 < 1$$

By the limit form of the ratio test, the Series is absolutely Convergent.

$\therefore$ It is Convergent.

(iii) $\frac{1^2 \cdot 2^2}{1!} + \frac{2^2 \cdot 3^2}{2!} + \frac{3^2 \cdot 4^2}{3!} + \cdots$

$\therefore \sum_{n=1}^{\infty} \frac{n^2 \cdot (n+1)^2}{n!}$

we apply the limit form of the ~~Com~~ ratio t

$$a_n = \frac{n^2 \cdot (n+1)^2}{n!}$$

$$a_{n+1} = \frac{(n+1)^2 \cdot (n+2)^2}{(n+1)!}$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(\cancel{n+1})^2 \cdot (n+2)^2}{(n+1)!} \times \frac{n!}{(n)^2 \cdot (\cancel{n+1})^2}\right|$$

$$= \left|\frac{(n+2)^2 \; \cancel{n!}}{(n+1) \; \cancel{n!} \, n^2}\right|$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+2)^2}{n^2(n+1)}$$

$$l = \lim_{n \to \infty} \left|\frac{a_{n+1}}{a_n}\right|$$

$$= \lim_{n \to \infty} \frac{(n+2)^2}{n^2(n+1)}$$

$$= \lim_{n \to \infty} \left[\frac{n^2 + 2n + 4}{n^3 + n^2}\right]$$

$$= \lim_{n \to \infty} \frac{\cancel{n^2}(1 + 2/n + 4/n^2)}{n^{\cancel{3}\,1}(1 + 1/n)}$$

$$= \lim_{n \to \infty} \frac{1}{n}$$

$$l = 0 < 1$$

$\therefore$ By the Limit form of the ratio test, the Series absolutely Convergent.

$\therefore$ It is Convergent //.

(04) (i) $\sum_{n=1}^{\alpha} \left(1+\frac{1}{n}\right)^{-n^2}$

Let $a_n = \dfrac{1}{(1+1/n)^{n^2}}$ $\forall n \in \mathbb{N}$

$$|a_n| = \left|\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}\right|$$

$$|a_n|^{1/n} = \frac{1^{1/n}}{\left(1+1/n\right)^{n^2 \times 1/n}}$$

$$|a_n|^{1/n} = \frac{1}{\left(1+\frac{1}{n}\right)^{n}}$$

$$\alpha = \lim\sup |a_n|^{1/n}$$

$$= \lim\sup \frac{1}{(1+1/n)^n}$$

$$\alpha = \lim_{n\to\alpha} \frac{1}{\left(1+1/n\right)^n}$$

Let $\lim_{n\to\alpha} \left(1+1/n\right)^n = Y$

$$\ln Y = \lim_{n\to\alpha} \ln\left(1+1/n\right)^n$$

$$= \lim_{n\to\alpha} n \ln\left(1+1/n\right)$$

$$= \lim_{n\to\alpha} \frac{\ln\left(1+1/n\right)}{1/n} \quad \frac{d/dn}{d/dn}$$

$$= \lim_{n\to\alpha} \frac{\left(\frac{\cancel{-1/n^2}}{1+1/n}\right)}{\left(\frac{\cancel{-1/n^2}}{1}\right)} \quad \frac{(\cancel{-n^2})}{(\cancel{-n^2})}$$

$$\ln Y = \lim_{n\to\alpha} \frac{1}{1+1/n}$$

$$\ln Y = 1$$

$$Y = e$$

$$\alpha = \frac{1}{y}$$

$$\alpha = \frac{1}{e} < 1$$

$\therefore$ By the root test, the Series is absolutely convergent.

$\therefore$ It is convergent.

(ii) $\sum_{n=1}^{\alpha} \frac{n^{n^2}}{(n+1)^{n^2}}$

Let $a_n = \frac{n^{n^2}}{(n+1)^{n^2}} \quad \forall n \in \mathbb{N}$

$$|a_n| = \left| \frac{n^{n^2}}{(n+1)^{n^2}} \right|$$

$$|a_n| = \left(\frac{n^{n^2}}{(n+1)^{n^2}} \right) \quad \forall n \in \mathbb{N}$$

$$|a_n|^{1/n} = \left(\frac{n}{n+1} \right)^{n^2 \times 1/n}$$

$$|a_n|^{1/n} = \left(\frac{n}{n+1} \right)^{n}$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$= \limsup \left[\left(\frac{n}{n+1} \right)^{n} \right]$$

$$\alpha = \lim_{n \to \alpha} \left[\left(\frac{n}{n+1} \right)^{n} \right]$$

$$= \lim_{n \to \alpha} \left[\frac{\cancel{n}}{\cancel{n}(1 + 1/n)} \right]^{n}$$

$$\alpha = \lim_{n \to \alpha} \frac{1}{(1 + 1/n)^{n}}$$

Let $\lim_{n \to \alpha} (1 + 1/n)^{n} = x$; $\alpha = \frac{1}{x}$

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$$\ln x = \ln \lim_{n\to\alpha} \left(1+\frac{1}{n}\right)^n$$

$$= \lim_{n\to\alpha} \ln \left(1+\frac{1}{n}\right)^n$$

$$= \lim_{n\to\alpha} n \ln \left(1+\frac{1}{n}\right)$$

$$= \lim_{n\to\alpha} \frac{\ln\left(1+\frac{1}{n}\right)}{\frac{1}{n}} \quad \frac{d/dn}{d/dn}$$

$$\ln x = \lim_{n\to\alpha} \left[\frac{\frac{-\cancel{1/n^2}}{1+1/n}}{\frac{-\cancel{1/n^2}}{1}}\right] \frac{(-\cancel{n^2})}{(-\cancel{n^2})}$$

$$\ln x = \lim_{n\to\alpha} \frac{1}{1+1/n}$$

$$\ln x = 1$$

$$x = e$$

$\therefore \ \alpha = \frac{1}{e} < 1$

$\therefore$ By the root test, the Series is absolutely Convergent.

$\therefore$ It is Convergent.

(iii) $\displaystyle\sum_{n=1}^{\alpha} \frac{1.3.5...(2n-1)\,x^{2n}}{2.4.6....2n\ 2^n}$

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Let $a_n = \dfrac{1.3.5...(2n-1)\ x^{2n}}{2.4.6...2n\ 2^n}$ $\forall\, n \in \mathbb{N}$

$$|a_n| = \left|\frac{1.3.5...(2n-1)\,x^{2n}}{2.4.6...\ 2n\ 2^n}\right|$$

$$|a_n| = \frac{1.3.5....(2n-1)\,x^{2n}}{2.4.6....\ 2n\ 2^n}$$

$$|a_n|^{1/n} = \left[\frac{1.3.5\ldots(2n-1)\,x^{2n}}{2.4.6\ldots 2n\,.\,2^n}\right]^{1/n}$$

~~Let a_n~~

$$|a_n|^{1/n} = \frac{1.3.5\ldots(2n-1)\,x^2}{(2.4.6\ldots 2n)\,2}$$

$\Rightarrow A = \{|a_1|^{1/1}, |a_2|^{1/2}, |a_3|^{1/3}, \ldots\}$

$A = \left\{\frac{1}{2}\left(\frac{x^2}{2}\right); \frac{1.3}{2.4}\left(\frac{x^2}{2}\right), \ldots\right\}$

$A = \left\{\frac{x^2}{4}, \frac{3x^2}{16}, \ldots\right\}$

$$\alpha = \lim\sup |a_n|^{1/n}$$

$$= \lim\sup\left[\frac{1.3.5\ldots(2n-1)\,x^2}{(2.4.6\ldots 2n)\,2.}\right]$$

$\sup A = \frac{x^2}{4}$

$$\alpha = \lim_{n\to\alpha} \frac{(2n-1)\,x^2}{2n\,.\,2.}$$

$$= \lim_{n\to\alpha} \frac{\cancel{n}\left[2-\frac{1}{n}\right]x^2}{2\cancel{n}\,.\,2}$$

$$= \frac{\cancel{2}x^2}{\cancel{4}2}$$

$$\alpha = \frac{x^2}{2} \quad ; \; x \in \mathbb{R}$$

when $x \leq 1$ ✗

$\therefore$ The Series $\sum_{n=1}^{\alpha} \frac{1.3.5\ldots(2n-1)\,x^{2n}}{2.4.6\ldots 2n\;2^n}$ is absolutely convergent.

when $x > 1$

$\therefore$ The series is divergent.

05 (i) $\sum_{n=2}^{\alpha} \frac{1}{n(\log n)^p}$

To apply the integral test, we need to consider the function $f(x) = \frac{1}{x(\log(x))^p}$ and determine whether its integral converges or diverges.

Now by integral test;

$$\lim_{n\to\alpha} \int_2^n \frac{1}{x(\log(x))^p} dx.$$

we can make a substitution $u = \log x$

$$u = \log x$$
$$du = \frac{1}{x} dx$$

$$\therefore \lim_{n\to\alpha} \int_2^n \frac{1}{x[\log(x)]^p} dx = \lim_{n\to\alpha} \int_2^n \frac{1}{u^p} du.$$

Now we know that the integral $\int_2^n \frac{1}{u^p} du$ Convergens.

if $p>1$ and diverges if $p \leq 1$

So, for our integral to Converge, we need $p>1$.

$\therefore$ The given series Converges if $p>1$ and diverges if $p\leq 1$

(iii) $\sum_{n=1}^{\infty} ne^{-n}$

Here $a_n = ne^{-n^2}$ $\forall n \in \mathbb{N}$

$f(x) = xe^{-x^2}$ $\forall n \in N$

(i) $f(x) > 0$ $\forall x \in [1, \alpha)$

(ii) $f(x)$ ✓

iii) $f(x)$ is Continuous

Now;

$$\lim_{n\to\alpha} \left[\int_1^n xe^{-x^2}\, dx\right]$$

$u = -x^2$

$\frac{du}{dx} = -2x$

$dx = \frac{du}{-2x}$

$$\lim_{n\to\alpha} \left[\int_1^n \cancel{x}.e^{u}.\frac{du}{-2\cancel{x}}\right]$$

$$\lim_{n\to\alpha} \left[\int_1^n -\frac{1}{2}e^{u}\, du\right]$$

$$\lim_{n\to\alpha} \left[-\frac{1}{2}\left\{e^{u}\right\}\right]_1^n$$

$$\lim_{n\to\alpha} \left[-\frac{1}{2}\left\{e^{-x^2}\right\}\right]_1^n$$

$$\lim_{n\to\alpha} \left[-\frac{1}{2}e^{-n^2} - \frac{1}{2}e^{-1^2}\right]$$

$$\lim_{n\to\alpha} \left\{-\frac{1}{2}\left[e^{-n^2} - e^{-1}\right]\right\}$$

$$\frac{-1}{2}\left[0 - 0.37\right]$$

$$0.185$$

$\therefore$ By the integral test the Series $\sum_{n=1}^{\alpha} ne^{-n^2}$ is Convergent.

06 (i) $\sum_{n=1}^{\alpha} \frac{(-1)^n \cos n\alpha}{n\sqrt{n}}$

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Let $a_n = \frac{(-1)^n \cos n\alpha}{n\sqrt{n}}$ $\forall n \in \mathbb{N}$

$|a_n| = \frac{\cos n\alpha}{n\sqrt{n}}$

Then $\sum_{n=1}^{\alpha} |a_n| = \sum_{n=1}^{\alpha} \frac{\cos n\alpha}{n\sqrt{n}} = \sum_{n=1}^{\alpha} \frac{\cos n\alpha}{n^{3/2}}$

Now, $p = 3/2 > 1$ ~~The Series is Convergent~~.

$\therefore$ we know that $1/n^{3/2}$ Series is absolutely Convergent.

$\left| \frac{\cos n\alpha}{n^{3/2}} \right| \leq \underbrace{\frac{1}{n^{3/2}}}_{b_n}$ → first, show that Σb_n is convergent or not

$\therefore$ By the Comparison test, $\sum_{n=1} \frac{(-1)^n \cos n\alpha}{n\sqrt{n}}$ is also absolutely Convergent.

$\therefore$ It is Convergent.

(ii) $\sum_{n=1}^{\alpha} \frac{\cos nx}{n^2}$

Let $a_n = \frac{\cos nx}{n^2}$ $\forall n \in \mathbb{N}$

$|a_n| = \left| \frac{\cos nx}{n^2} \right|$

$\sum_{n=1}^{\alpha} \cos nx = \cos x + \cos 2x + \dots + \cos nx$

$= \frac{\cos n\, x/2}{\cos x/2} \times \cos\left[x + (n-1)\frac{x}{2}\right]$

$\left| \sum_{n=1}^{\alpha} \cos nx \right| = \frac{|\cos n\, x/2|}{|\cos x/2|} \left[\cos x(n+1)\frac{x}{2}\right]$

$\left| \sum_{n=1}^{\alpha} \cos nx \right| < \frac{1}{|\cos x/2|} \times 1$

$$\left|\sum_{n=1}^{\alpha} \cos nx\right| < \frac{1}{|\cos x/2|}$$

This is bounded.

we know that $\sum_{n=1}^{\alpha} 1/n^2$ is convergent.

$\therefore \sum_{n=1}^{\alpha} \frac{\cos n\alpha}{n^2}$ is absoludely convergent.

(iv) $\sum_{n=1}^{\alpha} \frac{(-1)^n(n+2)}{2^n+5} \cos nx.$

Let $a_n = \frac{(-1)^n(n+2)}{2^n+5} \cos nx \quad ; n \in \mathbb{N}$

So, $|a_n| = \frac{(n+2) \cos nx}{2^n+5}$

Let $b_n = \frac{1}{n^2} \quad \forall n \in \mathbb{N}$

$$l = \lim_{n\to\alpha} \frac{a_n}{b_n}$$

$$l = \lim_{n\to\alpha} \left[\frac{(-1)^n(n+2)}{2^n+5} \times n^2\right]$$

$$l = \lim_{n\to\alpha} \frac{(-1)^n(n^3+2n^2)}{2^n+5}$$

$$l = \lim_{n\to\alpha} \frac{n^3/n^3 + 2n^2/n^3}{2^n/n^3 + 5/n^3}$$

* here, you can use ~~ratio~~ limit form of ratio test

$$l = \frac{1}{0}$$

l does not exit. ✗

$\therefore \sum_{n=1}^{\alpha} \frac{(-1)^n(n+2)}{2^n+5} \cos nx$ is divergent.

(07) ~~Since~~ $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}}$ are Convergent ?

Proof :-

Since $\sum_{n=1}^{\infty} a_n$ is Convergent $\lim_{n \to \infty} a_n = 0$

$\forall \varepsilon > 0$, $\exists$ an integer N

Such that $|a_n| < \varepsilon$ whenever $n \geq N$

Consider $\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{N} a_n^2 + \sum_{n=1}^{\infty} a_n^2$

$\sum_{n=1}^{N} a_n^2$ is Convergent because that Series is a finite Sum.

$\therefore \sum_{n=1}^{\infty} a_n^2$ is also convergent by the comparison test.

$\therefore \sum_{n=1}^{\infty} a_n^2$ is also Convergent.

Let Consider

$$\sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}} \leq \sqrt{\sum_{n=1}^{\infty} a_n} \times \sqrt{\sum_{n=1}^{\infty} a_{n+1}}$$

$\sqrt{\sum_{n=1}^{\infty} a_n}$ is Convergent. because, $\sum_{n=1}^{\infty} a_n$ is Convergent

$\therefore \sqrt{\sum_{n=1}^{\infty} a_{n+1}}$ is also Convergent. by the Comparison tes

$\therefore \sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}}$ is also Convergent.

(09) $\sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$

(i) Converges if $q > 1/2$;

7 $\sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$

In here;

$$a_n = \frac{1}{(n^2+n)^q}$$

Let;

$q = 1$

So, $a_n = \frac{1}{n^2+n}$ } this is not the proper way.

$$\frac{1}{n^2+n} < \frac{1}{n^2} = b_n$$

$\therefore$ Let $a_n = \frac{1}{n^2+n}$ and $b_n = \frac{1}{n^2}$

use p test

Then $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent.

$\therefore$ By the Comparison test $\sum_{n=1}^{\infty} a_n$ is Convergent.

$\therefore \sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$ is Convergent when $q > 1/2$.

(ii) Diverges if $q \leq 1/2$.

7 $a_n = \frac{1}{(n^2+n)^q}$

Let $q = 1/2$; So $a_n = \frac{1}{\sqrt{n^2+n}}$

$$\underbrace{\frac{1}{\sqrt{n^2+n}}}_{a_n} \leq \frac{1}{\sqrt{n^2}} = \underbrace{\frac{1}{n}}_{b_n}$$

use p test

Then $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} 1/n$ is divergent.

$\therefore$ By the Comparison test, $\sum_{n=1}^{\infty} a_n$ is divergent.

$\therefore \sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$ is divergent when $q \leq 1/2$ //

(10) (i) The Comparison test.

Proof :-

(a) Suppose that $\sum_{n=1}^{\infty} b_n$ Converge, where $b_n \geq 0$ $\forall n \in \mathbb{N}$

Let $b_n \geq a_n \geq 0$ $\forall n \in \mathbb{N}$

Take any $\varepsilon > 0$

Then by the Cauchy Criterion for Series

$\exists N(\varepsilon) \in \mathbb{N}$ Such that;

$|b_n - b_m| < \varepsilon$

$|b_{n+1} + b_{n+2} + b_{n+3} + \dots + b_m| < \varepsilon$ — (1)

whenever $m > n > N(\varepsilon)$

Note:- Cauchy Criterian for Series

* If Sequence (x_n) Converges then it Satisfies the Cauchy Criterion.

For $\varepsilon > 0$, $\exists N$ Such that,

$|x_n - x_m| < \varepsilon$ $\forall n, m \geq N$

Since $b_m \geq 0$ $\forall n \in \mathbb{N}$

$|b_{n+1} + b_{n+2} + \dots + b_m| = b_{n+1} + b_{n+2} + \dots + b_m$

Also Since $a_n \leq b_n$ $\forall n \in \mathbb{N}$ we have

$a_{n+1} + a_{n+2} + \dots + a_m \leq b_{n+1} + b_{n+2} + \dots + b_m$

we then have from (1)

$a_{n+1} + a_{n+2} + \dots + a_n < \varepsilon$ whenever $n, m > N(\varepsilon)$

Since; $a_n \geq 0$ $\forall n \in \mathbb{N}$

$|a_{n+1} + a_{n+2} + \dots + a_m| = a_{n+1} + a_{n+2} + \dots + a_m$

Hence;

$|a_{n+1} + a_{n+2} + \dots + a_m| < \varepsilon$ wheneve $m, n > N(\varepsilon)$

$\therefore$ by the Cauchy for Series $\sum_{n=1}^{\infty} a_n$ Convergent.

(b) Suppose that $\sum_{n=1}^{\infty} a_n$ is divergent we shall show that $\sum_{n=1}^{\infty} b_n$ divergent.

Let $S_n = \sum_{k=1}^{n} a_k \quad \forall n \in \mathbb{N}$

Since $a_n \geq 0 \quad \forall n \in \mathbb{N}$
(and divergent)

(ii) Limit form of the Comparison test.

(i) Converge Case;

Proof :-

Let $\sum b_n$ be a Convergent Series by the definition of Convergence, the Sequence of Partial Sum.

$S_n = \sum_{k=1}^{n} b_k$ Converges to Some finite limit.

Say S as n approaches infinitely

Now since a_n/b_n approaches L as n approaches infinity. for any $\varepsilon > 0$ $\exists$ an $N(\varepsilon)$ Such that,

$$\left| \frac{a_n}{b_n} - 1 \right| < \varepsilon \quad \forall n > N(\varepsilon)$$

$$-\varepsilon < \frac{a_n}{b_n} - 1 < \varepsilon$$

$$L - \varepsilon < \frac{a_n}{b_n} < L + \varepsilon \quad \forall n \geq N(\varepsilon)$$

$$(L-\varepsilon) b_n < a_n < (L+\varepsilon) b_n \quad \text{--- } (1)$$

let's take $(L-\varepsilon) b_n$ by c_n and $(L+\varepsilon) b_n$ by d_n

Since $\sum b_n$ is Convergent.

$\sum c_n$ x $\sum d_n$ are also Convergent.

Note:- $\sum_{n=1}^{\infty} a_n$ is Convergent, then $\sum_{n=1}^{\infty} c\, a_n$ is convergent where c is a Constant, ①⇒ $c_n \le b_n \le d_n$

Hence $\sum a_n$ also Converges by the Comparison test.

(ii) Divergence Case:

Let b_n be a divergent Series Similar to the Convergence Case, we have.

$(L-\varepsilon) b_n < a_n < (L+\varepsilon) b_n \quad \forall \ n_1 > N(\varepsilon)$

where $N(\varepsilon)$ is Some positive integer.

If $\sum a_n$ where to Converge the $\sum (L-\varepsilon) b_n$ and $\sum (L+\varepsilon) b_n$ would Converge,

$\Rightarrow \sum b_n$ Converges.

But this Contradicts the assumption, that b_n is divergent.

$\therefore$ if $\sum b_n$ is divergent.

$\Rightarrow \sum a_n$ must also be divergent.

(iii) Ration test.

Proof :-

a) Let $\left|\frac{a_{n+1}}{a_n}\right| \le r < 1$ — ① $\forall \ n \ge N$ where $a_n \ne 0 \ \forall \ n \in \mathbb{N}$

we must show that $\sum a_n$ is Convergent,

① → $|a_{n+1}| \le r|a_n| \quad \forall \ n \ge N$

$n=N \Rightarrow |a_{N+1}| \le r|a_N|$

$n=N+1 \Rightarrow |a_{N+2}| \le r|a_{N+2}| \Rightarrow |a_{N+1}| \le r^2 |a_N|$

$n=N+2 \Rightarrow |a_{N+3}| \le r|a_{N+3}| \Rightarrow |a_{N+3}| \le r^3|a_N|$

So, we have $|a_{N+j}| \le r^j |a_N| \quad \forall \ j = 1, 2, 3, \dots$

$\sum_{j=1}^{\infty} r^j |a_N| = |a_N| \sum_{n=1}^{\infty} r \cdot r^{j-1}$ is a geometric Series.

• geometric Series test ;

① Common ratio of $\sum_{n=1}^{\infty} r^{n-1}$

geometic test $|r| < 1 \Rightarrow$ Series is Convergent.

② $|r| \geq 1$ the geometric Series is divergent.

with Common ratio $r < 1$ and hence it is Convergent

Now by the Comparison test the Series $\sum_{j=1}^{\infty} |a_{N+j}|$ is Convergent.

$\therefore \sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{N} |a_n| + \sum_{j=N}^{\infty} |a_{n+j}|$ is Convergent.

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is Convergent.

b) Suppose that $\left|\frac{a_{n+1}}{a_n}\right| > r > 1 \quad \forall n \geq N$

$\Rightarrow |a_{n+1}| \geq r|a_n| \geq |a_n| \quad \forall n > N$

$\Rightarrow |a_N| \leq |a_{N+1}| \leq |a_{N+2}| \leq |a_{N+3}|$

$\Rightarrow (|a_{n+j}|)_j \quad ; j = 0, 1, 2, 3, \ldots$ is an increasing Sequence

$\lim_{n\to\infty} |a_n| \neq 0.$

$\lim_{n\to\infty} a_n \neq 0.$

$\Rightarrow$ The Series $\sum a_n$ is divergent //

v) Limit form of the ration test;

proof :-

Let $\lim_{n\to\alpha} \left|\frac{a_{n+1}}{a_n}\right| = l.$

$\Rightarrow$ for each $\varepsilon>0$ $\exists$ an $N(\varepsilon) \in \mathbb{N}$ Such that,

$$\left|\left|\frac{a_{n+1}}{a_n}\right| - l\right| < \varepsilon \quad \text{where} \quad n > N(\varepsilon) \rightarrow ①$$

i) Suppose that $l<1$

Let $\varepsilon = \frac{1-l}{2} > 0$.

$$\left(\frac{1-l}{2}\right) < \left|\frac{a_{n+1}}{a_n}\right| - l < \frac{1-l}{2}$$

$$\Rightarrow \left|\frac{a_{n+1}}{a_n}\right| < \frac{l+1}{2} \quad \forall\ n > N\left(\frac{1-l}{2}\right).$$

Let $r = \frac{l+1}{2} < 1$

Then we have

$$\left|\frac{a_{n+1}}{a_n}\right| < r \quad \forall\ n > N\left(\frac{1-l}{2}\right)$$

Now using Proof of ratio test we Conclude that $\sum a_n$ absolutely Converge

b) Let $l>1$ $\left|\left|\frac{a_{n+1}}{a_n}\right| - l\right| < \varepsilon$.

① $\Rightarrow$ $l-\varepsilon < \left|\frac{a_{n+1}}{a_n}\right| < l+\varepsilon$ — ② whenever $n > N(\varepsilon)$

Choose $\varepsilon = \frac{l-1}{2} > 0$ as $l>1$

$$\Rightarrow l-\varepsilon = l - \left(\frac{l-1}{2}\right) = \frac{l+1}{2} > 1$$

Then ② $\Rightarrow$

$$R < \left|\frac{a_{n+1}}{a_n}\right| \quad \text{whenever} \quad n > N\left(\frac{l-1}{2}\right) \quad \text{where} \quad R = \frac{l+1}{2} > 1$$

Now using Proof of ratio test. the Series $\sum a_n$ is divergent //.

(v) Alternative Series test;

Proof :- Let $S_n = \sum_{k=1}^{n} (-1)^{k-1} c_k \quad \forall n \in \mathbb{N}$.

$$\Rightarrow S_n = c_1 - c_2 + c_3 + \dots + c_{2n-1} - c_{2n}$$

$$= (c_1 - c_2) + (c_3 - c_4) + \dots + (c_{2n-1} - c_{2n})$$

$$= c_1 - [\underbrace{(c_2 - c_3)}_{\geq 0} + \underbrace{(c_4 - c_5)}_{\geq 0} + \dots + (c_{2n-2} - c_{2n-1}) + c_{2n}]$$

(decreasing sequence)

$$\Rightarrow 0 < c_1 - c_2 < S_{2n} < c_1 \quad \forall n \in \mathbb{N} \quad \text{— (1)}$$

$$S_{2n+2} - S_{2n} = S_{2n} + a_{n+2} - S_{2n} = a_{n+2} > 0 \quad \forall n \in \mathbb{N}$$

$\Rightarrow S_{2n}$ is increasing.

i.e, (S_{2n}) is increasing and bounded above.

$\therefore \lim_{n \to \infty} S_{2n} \in \mathbb{R}$

Let $\lim_{n \to \infty} S_{2n} = l_1$

Also $S_{2n+1} = \sum_{k=1}^{2n+1} (-1)^{k-1} c_k = S_{2n} + c_{2n+1}$ — (2)

$$= c_1 - c_2 + c_3 + c_4 + \dots - c_{2n} + c_{2n+1}$$

$$= c_1 - [\underbrace{(c_2 - c_3)}_{\geq 0} + \underbrace{(c_4 - c_5)}_{\geq 0} + \dots + (c_{2n} - c_{2n+1})]$$

$$\Rightarrow S_{2n} < S_{2n+1} < c_1 \quad \forall n \in \mathbb{N} \quad \text{— (3)}$$

By (1) and (3) we have

$$c_1 - c_2 < S_{2n} < S_{2n+1} < c_1 \quad \forall n \in \mathbb{N}$$

Also,

$$S_{2n+3} < S_{2n+1} \quad \forall n \in \mathbb{N}$$

$\Rightarrow (S_{2n+1})$ is decreasing sequence.

i.e; (S_{2n+1}) is decreasing and bounded above.

$$\therefore \lim_{n \to \alpha} S_{2n+1} \in \mathbb{R}$$

Let $\lim_{n \to \alpha} S_{2n+1} = l_2$

Letting $n \to \alpha$ in ② gives.

$$\lim_{n \to \alpha} S_{2n+1} = \lim_{n \to \alpha} S_{2n} + \lim_{n \to \alpha} C_{2n+1}$$

$$l_2 = l_1 + 0.$$

(as (C_n) is decreasing).

$\Rightarrow l_2 = l_1$

$\Rightarrow (S_n)$ is Convergent.

$\Rightarrow \sum_{n=1}^{\alpha} (-1)^{n-1} C_n$ is Convergent.